Method of Differentiation

131. Let $f : R \to R$ be a twice differentiable function such $(\sin x \cos y)(f(2x + 2y) - f(2x - 2y)) = (\cos x \sin y)(f(2x + 2y) + f(2x - 2y))$, for all $x, y \in R$.

If $f'(0) = \frac{1}{2}$, then the value of $24f''\left(\frac{5\pi}{3}\right)$ is :

माना एक फलन $f : R \to R$ दो बार अवकलनीय है तथा सभी $x, y \in R$ के लिए

$(\sin x \cos y)(f(2x + 2y) - f(2x - 2y)) = (\cos x \sin y)(f(2x + 2y) + f(2x - 2y))$ है। यदि $f'(0) = \frac{1}{2}$ है, तो $24f''\left(\frac{5\pi}{3}\right)$ का मान है :

(A) 2 (B) –2 (C) 3 (D) –3

Ans. D

Sol. $(\sin x \cos y)(f(2x + 2y) - f(2x - 2y)) = (\cos x \sin y)(f(2x + 2y) + f(2x - 2y))$

$f(2x + 2y)(\sin(x - y)) = f(2x - 2y)\sin(x + y)$

$$\frac{f(2x+2y)}{\sin(x+y)} = \frac{f(2x-2y)}{\sin(x-y)}$$

Put $2x + 2y = m$, $2x - 2y = n$

$$\frac{f(m)}{\sin\left(\frac{m}{2}\right)} = \frac{f(n)}{\sin\left(\frac{n}{2}\right)} = K$$

$$\Rightarrow f(m) = K\sin\left(\frac{m}{2}\right)$$

$$\therefore f(x) = K\sin\left(\frac{x}{2}\right)$$

Put $x = 0$; $\frac{1}{2} = \frac{K}{2}$ $\Rightarrow K = 1$

$$f'(x) = \frac{1}{2}\cos\frac{x}{2}$$

$$f''(x) = -\frac{1}{4}\sin\frac{x}{2}$$

$$4f''\left(\frac{5\pi}{3}\right) = \left(-\frac{1}{4}\sin\left(\frac{5\pi}{6}\right)\right)24$$

$= -\frac{24}{8} = -3$

132. If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in R$, then $\frac{d^2y}{dx^2} + y$ is equal to :

यदि $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in R$ है, तो $\frac{d^2y}{dx^2} + y$ बराबर है :

(A) –1 (B) 28 (C) 27 (D) 1

Ans. A

Sol. $C_3 \to C_3 - C_1$

$$y(x) = \begin{vmatrix} \sin x & \cos x & 1 + \cos x \\ 27 & 28 & 0 \\ 1 & 1 & 0 \end{vmatrix}$$

$y(x) = -(1 + \cos x)$

$\frac{dy}{dx} = \sin x \qquad \Rightarrow \frac{d^2y}{dx^2} = \cos x$

$\frac{d^2y}{dx^2} + y = -1$

133. Let $f(x) = x^5 + 2x^3 + 3x + 1$, $x \in \mathbf{R}$, and $g(x)$ be a function such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. the $\frac{g(7)}{g'(7)}$ is equal to :

माना $f(x) = x^5 + 2x^3 + 3x + 1$, $x \in \mathbf{R}$ है, तथा एक फलन $g(x)$ इस प्रकार है कि $g(f(x)) = x \ \forall \ x \in R$ है। तो $\frac{g(7)}{g'(7)}$ बराबर है :

(A) 42 (B) 7 (C) 14 (D) 1

Ans. C

Sol. $f(x) = x^5 + 2x^3 + 3x + 1$

$f'(x) = 5x^4 + 6x^2 + 3$

$f'(1) = 5 + 6 + 3 = 14$

$g(f(x)) = x$

$g'(f(x))f'(x) = 1$

for $f(x) = 7$

$\Rightarrow x^5 + 2x^3 + 3x + 1 = 7$

$\Rightarrow x = 1$

$g'(7)\, f'(1) = 1 \Rightarrow g'(7) = \frac{1}{f'(1)} = \frac{1}{14}$

$x = 1, f(x) = 7 \Rightarrow g(7) = 1$

$\frac{g(7)}{g'(7)} = \frac{1}{1/14} = 14$

134. If $y(\theta) = \frac{2\cos\theta + \cos 2\theta}{\cos 3\theta + 4\cos 2\theta + 5\cos\theta + 2}$, then at $\theta = \frac{\pi}{2}$, $y'' + y' + y$ is equal to :

यदि $y(\theta) = \frac{2\cos\theta + \cos 2\theta}{\cos 3\theta + 4\cos 2\theta + 5\cos\theta + 2}$ है, तो $\theta = \frac{\pi}{2}$, $y'' + y' + y$ बराबर है :

(A) 2 (B) 1/2 (C) 1 (D) 3/2

Ans. A

Sol. $y = \frac{2\cos\theta + 2\cos^2\theta - 1}{4\cos^3\theta - 3\cos\theta + 8\cos^2\theta - 4 + 5\cos\theta + 2}$

$y = \frac{\left(2\cos^2\theta + 2\cos\theta - 1\right)}{\left(2\cos^2\theta + 2\cos\theta - 1\right)\left(2\cos\theta + 2\right)}$

$y = \frac{1}{2}\left(\frac{1}{1+\cos\theta}\right)$

$\Rightarrow \theta = \frac{\pi}{2} \qquad y = \frac{1}{2}$

$y' = \frac{1}{2}\left(\frac{-1}{(1+\cos\theta)^2} \times (-\sin\theta)\right)$

$\Rightarrow \theta = \frac{\pi}{2} \qquad y' = \frac{1}{2}$

$$y'' = \frac{1}{2}\left[\frac{\cos\theta(1+\cos\theta)^2 - \sin\theta(2)(1+\cos\theta)(-\sin\theta)}{(1+\cos\theta)^4}\right]$$

$\Rightarrow \theta = \frac{\pi}{2}$ $\quad$ y'' = 1

135. Suppose for a differentiable function h, h(0)=0, h(1)=1 and $h'(0) = h'(1) = 2$. If $g(x) = h\left(e^x\right)e^{h(x)}$, then $g'(0)$ is equal to :

मान लीजिए कि अवकलनीय फलन h के लिए, h(0)=0, h(1)=1 तथा $h'(0) = h'(1) = 2$ हैं। यदि $g(x) = h\left(e^x\right)e^{h(x)}$, तो $g'(0)$ बराबर है :

(A) 3 (B) 8 (C) 4 (D) 5

Ans. C

Sol. $g(x) = h(e^x).e^{h(x)}$

$g'(x) = h(e^x).e^{h(x)}.h'(x) + e^{h(x)}h'(e^x).e^x$

$g'(0) = h(1)^{eh(0)}h'(0) + e^{h(0)}h'(1)$

$= 2 + 2 = 4$

136. Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that f(1) = 40, f '(1) = 2 and f '' (1) = 4. Then $a^2 + b^2 + c^2$ is equal to :

माना $f(x) = ax^3 + bx^2 + cx + 41$ इस प्रकार है कि f(1) = 40, f '(1) = 2 तथा f '' (1) = 4 है। तो $a^2 + b^2 + c^2$ बराबर है :

(A) 73 (B) 62 (C) 51 (D) 54

Ans. C

Sol. $f(x) = ax^3 + bx^2 + cx + 41$

$f'(x) = 3ax^2 + 2bx + c$

$\Rightarrow f'(1) = 3a + 2b + c = 2$ .....(1)

f'' (n) = 6ax + 2b

$\Rightarrow f''(1) = 6a + 2b = 4$

$3a + b = 2$ ...(2)

(A) – (2) $\Rightarrow b + c = 0$ ...(3)

f(1) = 40

a + b + c + 41 = 40

use (3)

$a + 41 = 40$

by (2)

$-3 + b = 2 \Rightarrow b = 5$ & $c = -5$

$a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$

137. If $\log_e y = 3\sin^{-1}x$, then $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ is equal to :

यदि $\log_e y = 3\sin^{-1}x$ है, तो $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ बराबर है :

(A) $9e^{\frac{\pi}{6}}$ (B) $3e^{\frac{\pi}{2}}$ (C) $3e^{\frac{\pi}{6}}$ (D) $9e^{\frac{\pi}{2}}$

Ans. D

Sol. $\log_e y = 3\sin^{-1}x$

$y = e^{3\sin^{-1}x}$

$\frac{dy}{dx} = e^{3\sin^{-1}x} \cdot \frac{3}{\sqrt{1-x^2}}$

$\sqrt{1-x^2}\frac{dy}{dx} = 3y$

Again differentiate

$\sqrt{1-x^2} \cdot y'' - \frac{2x}{2\sqrt{1-x^2}}y' = 3y'$

$(1-x)^2 y'' - xy' = 3y'\left(\sqrt{1-x^2}\right)$

So value of $3y'\left(\sqrt{1-x^2}\right)$

at $x = \frac{1}{2}$

$3 \cdot \frac{3}{\sqrt{1-x^2}} e^{\sin^{-1}x}\left(\sqrt{1-x^2}\right)$

$= 9e^{3\frac{\pi}{6}} = 9e^{\frac{\pi}{2}}$

138. Suppose $f(x)=\frac{\left(2^x+2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2-x+1\right)}}{\left(7x^2+3x+1\right)^3}$. Then the value of f'(0) is equal to :

माना $f(x)=\frac{\left(2^x+2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2-x+1\right)}}{\left(7x^2+3x+1\right)^3}$ है। तो f '(0) का मान बराबर है :

(A) $\frac{\pi}{2}$ (B) π (C) 0 (D) $\sqrt{\pi}$

Ans. D

Sol. Let $f(x)=\tan x\, g(x)$

where $g(x)=\frac{\left(2^x+2^{-x}\right)\sqrt{\tan^{-1}\left(x^2-x+1\right)}}{\left(7x^2+3x+1\right)^3}$

$\Rightarrow f'(x)=\sec^2 x g(x)+\tan x g'(x)$

$f'(0)=\sec^2(0)g(0)+\tan 0\, g'(0)$

$=g(0)$

$g(0)=\frac{\left(2^0+2^{-0}\right)\sqrt{\tan^{-1}\left(0^2-0+1\right)}}{\left(7(0)^2+3(0)+1\right)^3}$

$=2\sqrt{\tan^{-1}1}=2\sqrt{\frac{\pi}{4}}=\sqrt{\pi}$

139. Let $y=\log_e\left(\frac{1-x^2}{1+x^2}\right), -1<x<1$. Then at $x=\frac{1}{2}$, the value of $225(y'-y'')$ is equal to :

माना $y=\log_e\left(\frac{1-x^2}{1+x^2}\right), -1<x<1$ है। तो $x=\frac{1}{2}$ पर $225(y'-y'')$ का मान है :

(A) 736 (B) 746 (C) 732 (D) 742

Ans. A

Sol. $y=\log_e\left(\frac{1-x^2}{1+x^2}\right)$

$$\frac{dy}{dx} = y' = \frac{-4x}{1-x^4}$$

Again $\frac{d^2y}{dx^2} = y'' = \frac{-4\left(1+3x^4\right)}{\left(1-x^4\right)^2}$

Again $y' - y'' = \frac{-4x}{1-x^4} + \frac{4\left(1+3x^4\right)}{\left(1-x^4\right)^2}$

at $x = \frac{1}{2}$

$$y' - y'' = \frac{736}{225}$$

Thus $225\left(y' - y''\right) = 225 \times \frac{736}{225} = 736$

140. If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to :

यदि $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ है, तो $\frac{1}{5}f'(0)$ बराबर है :

(A) 2 (B) 6 (C) 0 (D) 1

Ans. C

Sol. $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$

$$f'(x) = \begin{vmatrix} 8\cos^3 x(-\sin x) & 2\sin^4 x & 3+\sin^2 2x \\ 8\cos^3 x(-\sin x) & 2\sin^4 x & \sin^2 2x \\ 8\cos^3 x(-\sin x) & 3+2\sin^4 x & \sin^2 2x \end{vmatrix} + \begin{vmatrix} 2\cos^4 x & 8\sin^3 x\cos x & 3+\sin^2 2x \\ 3+2\cos^4 x & 8\sin^3 x\cos x & \sin^2 2x \\ 2\cos^4 x & 8\sin^3 x\cos x & \sin^2 2x \end{vmatrix} +$$

$$\begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 4\sin 2x\cos 2x \\ 3+2\cos^4 x & 2\sin^4 x & 4\sin 2x\cos 2x \\ 2\cos^4 x & 3+2\sin^4 x & 4\sin 2x\cos 2x \end{vmatrix}$$

$$f'(0)=\begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}+\begin{vmatrix} 2 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 0 & 0 \end{vmatrix}+\begin{vmatrix} 2 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 0 & 0 \end{vmatrix}$$

$f'(0)=0$

$\therefore \frac{1}{5}f'(0)=0$

141. If $f(x)=\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ for all $x \in R$, then $2f(0)+f'(0)$ is equal to :

यदि सभी $x \in R$ के लिए $f(x)=\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ हैं, तो $2f(0)+f'(0)$ बराबर है :

(A) 48 (B) 18 (C) 24 (D) 42

Ans. D

Sol. $f(x)=\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$

$$f'(x)=\begin{vmatrix} 3x^2 & 4x & 3 \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-x \end{vmatrix}+\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 6x & 2 & 3x^2 \\ x^3-x & 4 & x^2-2 \end{vmatrix}+\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ 3x^2-1 & 0 & 2x \end{vmatrix}$$

$$2f(0)+f'(0)=2\begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix}+\begin{vmatrix} 0 & 0 & 3 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix}+\begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & -2 \end{vmatrix}+\begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ -1 & 0 & 0 \end{vmatrix}$$

$= 24 + 24 + 0 - 6$

= 42

142. Let g(x) be a linear function and $f(x)=\begin{cases} g(x) & , \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , \quad x > 0 \end{cases}$, is continuous at x = 0. If f′(1) = f(–1), then the value of g(3) is :

माना g(x) एक रैखिक फलन है तथा $f(x)=\begin{cases} g(x) & , \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , \quad x > 0 \end{cases}$, x = 0 पर संतत है। यदि f′(1) = f(–1) है, तो g(3) का मान है :

(A) $\log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$ (B) $\log_e\left(\frac{4}{9}\right)-1$ (C) $\frac{1}{3}\log_e\left(\frac{4}{9}\right)+1$ (D) $\frac{1}{3}\log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$

Ans. A

Sol. Let g(x) = ax + b

f(x) is continuous at x = 0

$\therefore \underset{x\to 0^+}{Lt}\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}=\left(\frac{1}{2}\right)^{\infty}=0$

$\underset{x\to 0^-}{Lt} g(x)=\underset{x\to 0^-}{Lt} ax+b=b$

So, b = 0

Now, $f(x)=\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}$

$\ell n f(x)=\frac{1}{x}\left[\ell n(1+x)-\ell n(2+x)\right]$

$\frac{1}{f(x)}\cdot f'(x)=\frac{x\left[\frac{1}{1+x}-\frac{1}{2+x}\right]-\ell n\left(\frac{1+x}{2+x}\right)}{x^2}$

$$f'(1) = f(1)\left[\frac{\left(\frac{1}{2}-\frac{1}{3}\right)-\ell n\left(\frac{2}{3}\right)}{1}\right]$$

$$f'(1) = \frac{2}{3}\left[\frac{1}{6}-\ell n\frac{2}{3}\right] \quad ...(1)$$

Also f(–1) a(–1) + b

$f(-1) = -a$ ...(2)

$\therefore$ from (1) and (2)

$$-a = \frac{1}{9}-\frac{2}{3}\ell n\frac{2}{3}$$

$$a = \frac{2}{3}\ell n\left(\frac{2}{3}\right)-\frac{1}{9}$$

$\therefore g(x) = ax$

$$g(3) = 3\left(\frac{2}{3}\ell n\left(\frac{2}{3}\right)-\frac{1}{9}\right)$$

$$g(3) = 2\ell n\left(\frac{2}{3}\right)-\frac{1}{3}$$

$$g(3) = \ell n\frac{4}{9}-\frac{1}{3}$$

$$g(3) = \ell n\frac{4}{9}-\ell ne^{\frac{1}{3}}$$

$$g(3) = \ell n\left(\frac{4}{9\cdot e^{\frac{1}{3}}}\right)$$

$$g(3) = \log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$$

143. Let $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}, x \in [0,\pi] - \left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to :

यदि $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}, x \in [0,\pi] - \left\{\frac{\pi}{4}\right\}$ है, तो $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ बराबर है :

(A) $\frac{2}{9}$ (B) $\frac{-2}{3}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{-1}{3\sqrt{3}}$

Ans. A

Sol. $f(x)=\sqrt{2}\frac{\left(\frac{1}{\sqrt{2}}\sin x+\frac{1}{\sqrt{2}}\cos x\right)-1}{\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x-\frac{1}{\sqrt{2}}\cos x\right)}=\frac{\sin(x+\pi/4)-1}{\sin(x-\pi/4)}$

$$f'(x)=\frac{\cos(x+\pi/4)\sin(x-\pi/4)-\cos(x-\pi/4)(\sin(x+\pi/4)-1)}{\sin^2(x-\pi/4)}$$

$$f'(x)=\frac{\cos(x-\pi/4)-1}{\sin^2(x-\pi/4)}=\frac{-(1-\cos(x-\pi/4))}{1-\cos^2(x-\pi/4)}$$

$$f'(x)=-\frac{1}{1+\cos(x-\pi/4)}$$

$$=f''(x)=-\frac{\sin(x-\pi/4)}{(1+\cos(x-\pi/4))^2}$$

$$f(7\pi/12)=-\frac{1}{\sqrt{3}}$$

$$f''(7\pi/12)=-\frac{2\sqrt{3}}{9}$$

$$f(7\pi/12)f''(7\pi/12)=\frac{2}{9}$$

144. The value of $\lim\limits_{x\to0}2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____.

$\lim\limits_{x\to0}2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ का मान होगा _____

Ans. 55

Sol. $\lim\limits_{x\to0}\frac{2}{x^2}\left(1-\left(1-\frac{x^2}{2}\right)\left(1-\frac{2^2x^2}{2}\right)^{\frac{1}{2}}\left(1-\frac{3^2x^2}{2}\right)^{\frac{1}{3}}.....\left(1-\frac{10^2x^2}{2}\right)^{\frac{1}{10}}\right)$

$$\lim_{x\to0}\frac{2}{x^2}\left(1-\left(1-\frac{x^2}{2}\right)\left(1-\frac{2^2x^2}{2.2}\right)\left(1-\frac{3^2\cdot x^2}{3.2}\right).....\left(1-\frac{10^2x^2}{2.10}\right)\right)$$

$$= \lim_{x\to 0} \frac{2}{x^2}\left(1-\left(1-\frac{x^2}{2}-\frac{2x^2}{2}-\frac{3x^2}{2}\ldots\ldots\ldots-\frac{10x^2}{2}\right)\right)$$

$$= \left(\frac{2}{2}(1+2+3+\ldots\ldots+10)\right) = \frac{10(10+1)}{2} = 55$$

145. Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3), x \in R$. Then $f'(10)$ is equal to _____.

माना $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3), x \in R$ है। तो $f'(10)$ बराबर है _____।

Ans. 202

Sol. $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$

$f'(x) = 3x^2 + 2x f'(1) + f''(2)$

$f''(x) = 6x + 2f'(1)$

$f'''(x) = 6$

$\Rightarrow f'''(3) = 6$

$f''(2) = 12 + 2f'(1)$ _____(1)

$f'(1) = 3 + 2f'(1) + f''(2)$ _____(2)

adding (1) and (2)

$0 = 15 + 3f'(x) \Rightarrow f'(1) = -5$

$f''(2) = 12 - 10 = 2$

Now

$f'(10) = 300 + 20(-5) + 2$

$= 300 - 100 + 2 = 202$

146. If $y = \frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$, then $96\, y'\left(\frac{\pi}{6}\right)$ is equal to _____.

यदि $y = \frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$ है, तो $96\, y'\left(\frac{\pi}{6}\right)$ बराबर है _____

Ans. 105

Sol. $y = \frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(x\sqrt{x}-1\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2\cdot x - 5\right)\cos^3 x$

$$y=\frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(\sqrt{x}\right)^3-1^3\right)}{x\sqrt{x}+x+\sqrt{x}}+\frac{1}{15}\left[3\cos^5 x-5\cos^3 x\right]$$

$$y=\frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(\sqrt{x}-1\right)\left(x+\sqrt{x}+1\right)}{\sqrt{x}\left(x+\sqrt{x}+1\right)}+\frac{1}{15}\left[3\cos^5 x-5\cos^3 x\right]$$

$$y=(x-1)+\frac{1}{15}\left[3\cos^5 x-5\cos^3 x\right]$$

Now, diff w.r.t x

$$y'=1-0+\frac{1}{15}\left[15\cos^4 x\cdot(-\sin x)-15\cos^2 x(-\sin x)\right]$$

$$y'=1-\sin x\cdot\left[\cos^4 x-\cos^2 x\right]$$

$$y'\left(\frac{\pi}{6}\right)=1-\sin\frac{\pi}{6}\left[\cos^4\frac{\pi}{6}-\cos^2\frac{\pi}{6}\right]$$

$$=1-\frac{1}{2}\left[\frac{9}{16}-\frac{3}{4}\right]$$

$$=1-\frac{1}{2}\left[\frac{9-12}{16}\right]\Rightarrow 1+\frac{3}{32}\Rightarrow\frac{35}{32}$$

$$96.y'\left(\frac{\pi}{6}\right)=96\times\frac{35}{32}\Rightarrow 105$$